

Формулы

$$1) v_x = v_0 \cos \alpha$$

$$3) v_y = v_0 \sin \alpha - gt$$

$$2) x = v_0 \cos \alpha \cdot t$$

$$4) y = v_0 \sin \alpha \cdot t - \frac{gt^2}{2}$$

$$5) x = S ; y = -h$$

$$6) v = \sqrt{v_x^2 + v_y^2}$$

Вывод формул:

из (4) и (5):

$$\begin{aligned} y &= -h \\ y &= v_0 \sin \alpha \cdot t - \frac{gt^2}{2} \end{aligned} \Rightarrow$$

$$\Rightarrow -h = v_0 \sin \alpha \cdot t - \frac{gt^2}{2}$$

$$\frac{gt^2}{2} - v_0 \sin \alpha \cdot t - h = 0$$

$$D = v_0^2 \sin^2 \alpha - 4 \cdot \frac{g}{2} \cdot (-h) = v_0^2 \sin^2 \alpha + 2gh$$

$$t_1 = \frac{v_0 \sin \alpha + \sqrt{v_0^2 \sin^2 \alpha + 2gh}}{g}$$

$$t_2 = \frac{v_0 \sin \alpha - \sqrt{v_0^2 \sin^2 \alpha + 2gh}}{g} \quad | < 0 \Rightarrow$$

$\Rightarrow$ не подходит.

7) подставляем t_1 в (2) и (3) с учетом (5):

$$\begin{aligned} x &= v_0 \cos \alpha \cdot t \\ x &= S \end{aligned} \Rightarrow S = v_0 \cos \alpha \cdot t, \text{ подставляем } t_1$$

$$7) \Rightarrow S = \frac{v_0^2 \sin \alpha \cos \alpha + v_0 \cos \alpha \sqrt{v_0^2 \sin^2 \alpha + 2gh}}{g}$$

$$8) v_y = v_0 \sin \alpha - gt, \text{ nogeturum } t, \Rightarrow$$

$$\Rightarrow v_y = v_0 \sin \alpha - g \cdot \frac{v_0 \sin \alpha + \sqrt{v_0^2 \sin^2 \alpha + 2gh}}{g} =$$

$$= v_0 \sin \alpha - v_0 \sin \alpha + \sqrt{v_0^2 \sin^2 \alpha + 2gh} =$$

$$= \sqrt{v_0^2 \sin^2 \alpha + 2gh}$$

$$9) \text{ u) (6) e yzērom (1) u (8):}$$

$$v = \sqrt{v_x^2 + v_y^2} \quad v_x = v_0 \cos \alpha$$

$$v_y = -\sqrt{v_0^2 \sin^2 \alpha + 2gh}$$

$$v = \sqrt{v_0^2 \cos^2 \alpha + v_0^2 \sin^2 \alpha + 2gh} =$$

$$= \sqrt{v_0^2 (\cos^2 \alpha + \sin^2 \alpha) + 2gh} =$$

$$= \sqrt{v_0^2 + 2gh}$$

$$10) \text{ u) (2) naugēm } t \text{ u nogeturum } t(4)$$

$$x = v_0 \cos \alpha \cdot t \Rightarrow t = \frac{x}{v_0 \cos \alpha}$$

$$y = v_0 \sin \alpha \cdot \frac{x}{v_0 \cos \alpha} - \frac{g \cdot x^2}{2 v_0^2 \cos^2 \alpha} =$$

$$= \tan \alpha \cdot x - \frac{g \cdot x^2}{2 v_0^2 \cos^2 \alpha}$$